

Network Layer (2)

COMP90007 Internet Technologies

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Types of Services

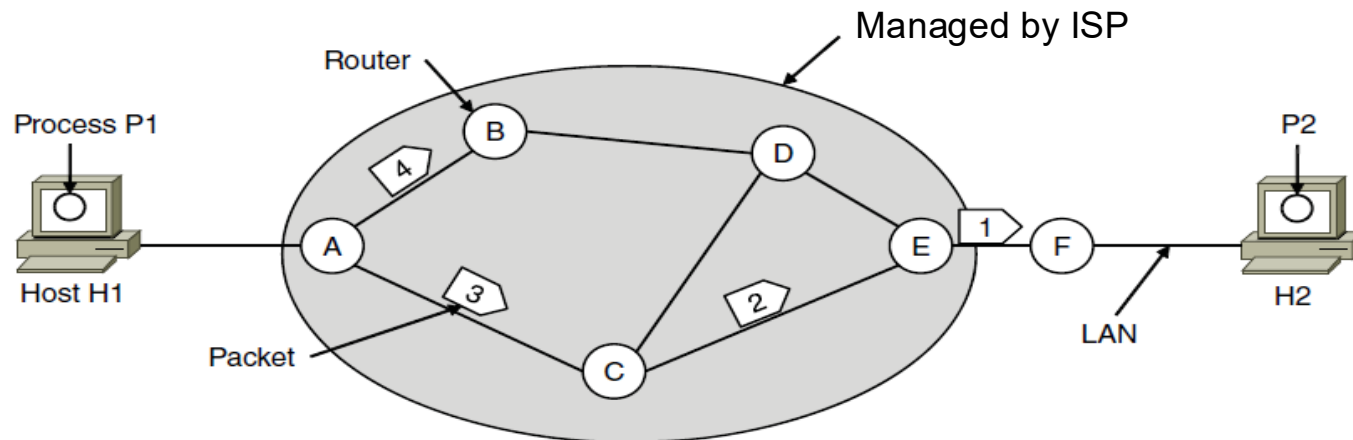
- **Connectionless:** Packets are injected into subnet **individually** and routed **independently** to the destination
 - Internet community: move packets in a potentially unreliable subnet
 - QoS is not easily implemented, e.g., flow and error control done by other layers
- **Connection-oriented:** Packets travelling to the destination following the **same route**
 - Telecommunication community: guarantee reliability
 - QoS is important

Routing within a Datagram Subnet

■ Connectionless - post office model

Packets are routed individually based on the destination addresses

- Packets can take different paths
- Example: P1 sends a long message to P2



Routing table

Routing algorithm –
manages the routing table

A's table (initially)

A	–
B	B
C	C
D	B
E	C
F	C

Dest. Line

A's table (later)

A	–
B	B
C	C
D	B
E	B
F	B

C's Table

A	A
B	A
C	–
D	E
E	E
F	E

E's Table

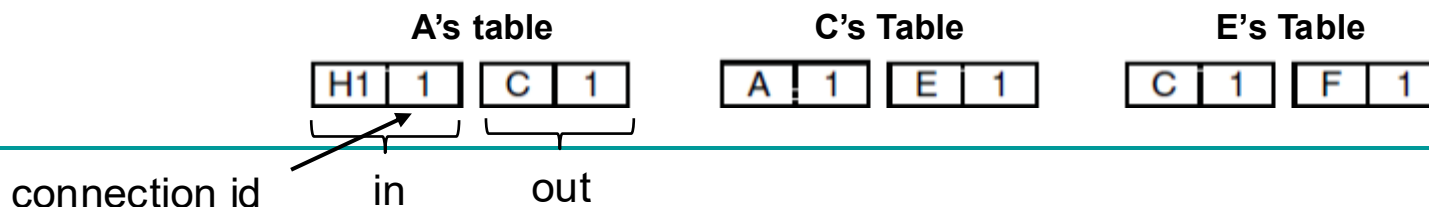
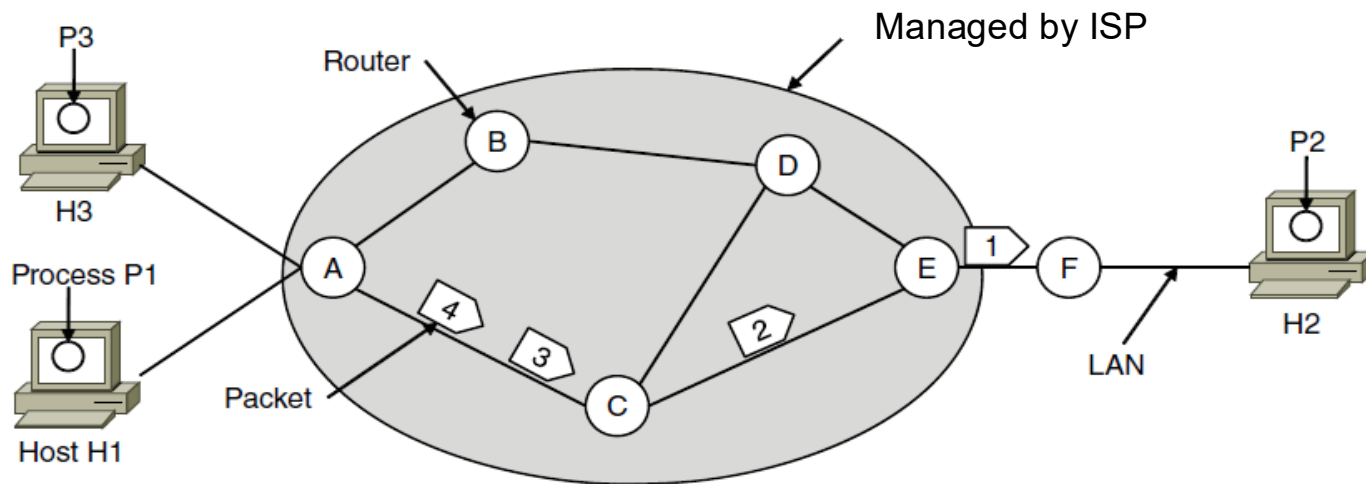
A	C
B	D
C	C
D	D
E	–
F	F

Routing within a Virtual-Circuit Subnet

■ Connection-oriented - telephone network model

Packets are routed through virtual circuits based on connection id

- ❑ Packets take the same path to avoid choosing a new path for every packet
- ❑ Example: Multiprotocol Label Switching Network (MPLS)



Datagram vs. Virtual-Circuit Subnets (1)

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC

Datagram vs. Virtual-Circuit Subnets (2)

■ Time consumption

- ❑ **VC:** requires setup time and resources, but packet transmission is very fast after that
- ❑ **Datagram:** more address parsing time due to complicated lookup procedure

■ Memory of router

- ❑ **VC:** requires one entry per virtual circuit
- ❑ **Datagram:** requires large tables of every possible destination

Datagram vs. Virtual-Circuit Subnets (3)

■ Longevity

- **VC:** suitable for repeating and long-running uses, e.g. Permanent VC's
- **Datagram:** suitable for frequently changed connections

■ Vulnerability

- **VC:** particularly vulnerable to hardware/software crashes, all VC's aborted and no traffic until they are rebuilt
- **Datagram:** can use an alternative route

Different Networks

- Service offered: connectionless vs. connection-oriented
- Packet size: different maximum sizes
- Addressing: different lengths, flat or hierarchical
- Quality of service: present or absent
- Security: privacy rules, encryption

Outline

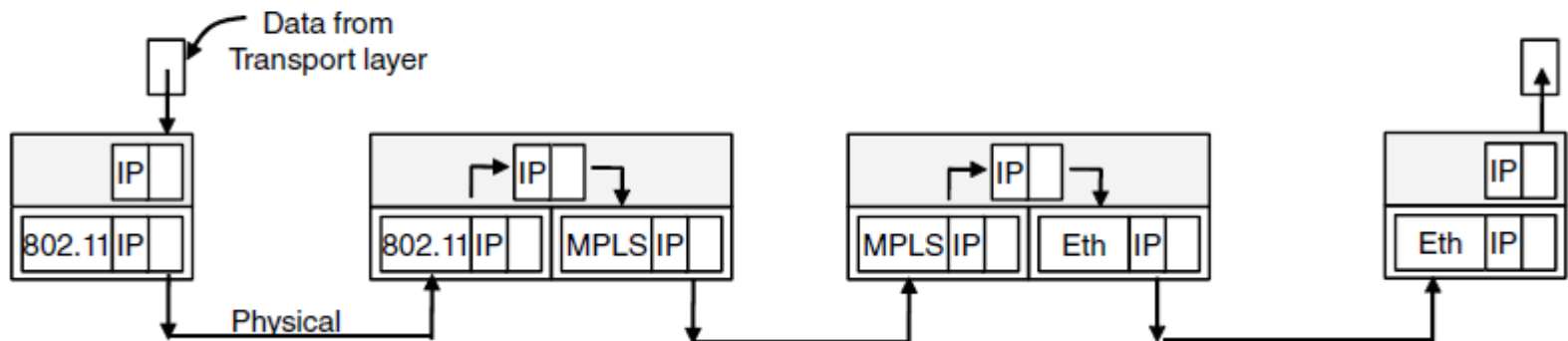
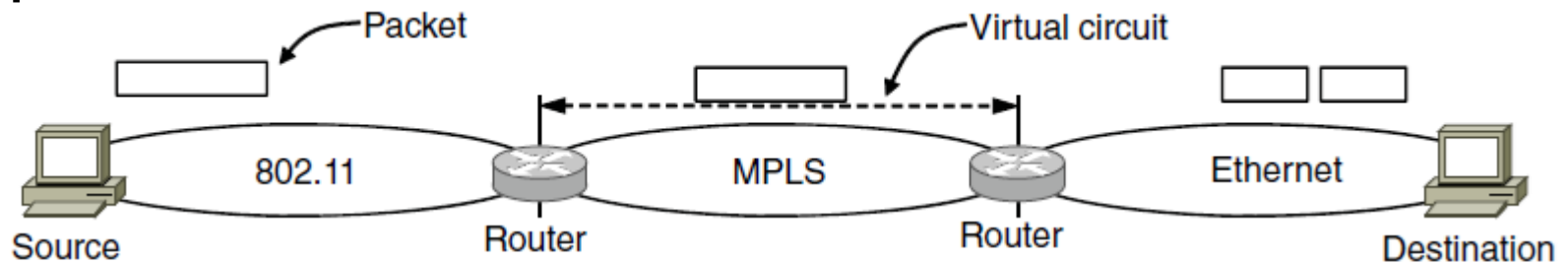
- Network layer on the Internet
- Types of services
- Internetworking
 - Tunneling
 - Fragmentation
 - Path MTU discovery
- Internet Protocol
 - Addressing
 - Subnetting
- Routing algorithms

Internetworking

- Internetworking joins multiple, different networks into a single larger network
- Challenges in connecting networks:
 - Different network types and protocols
 - Incompatibilities across hardware and software technologies
 - Different design goals and motivations behind network configurations

Internetworking

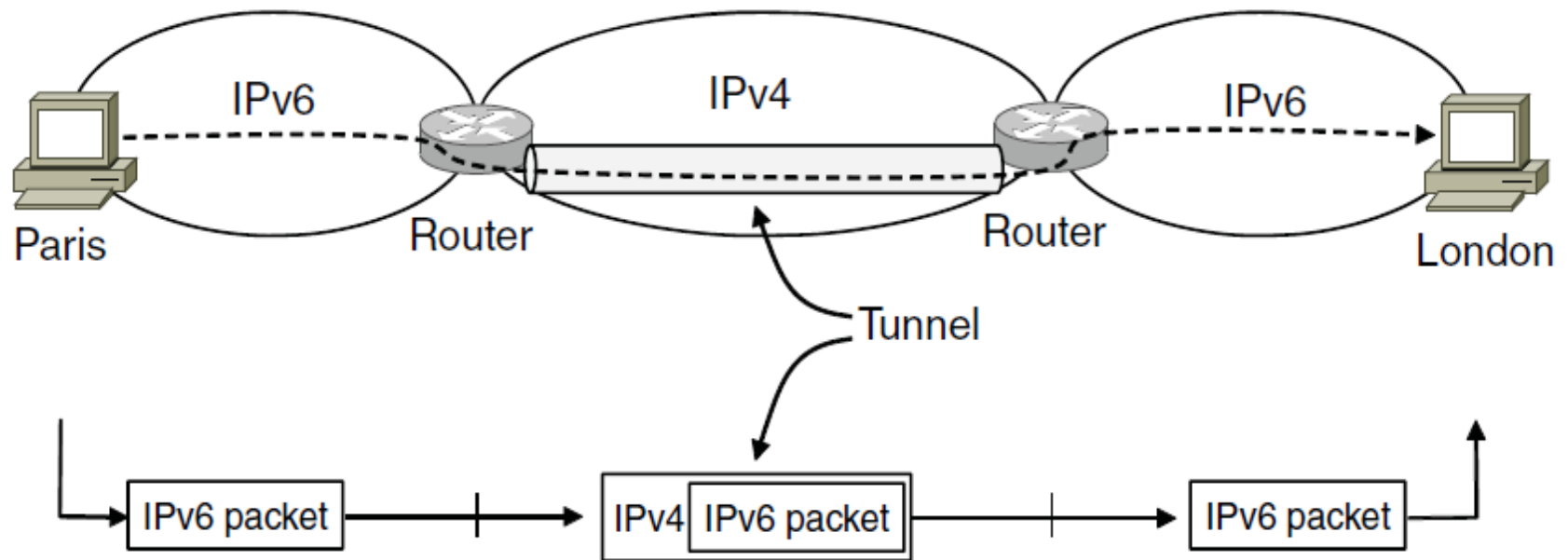
- Internetworking using a common network layer protocol IP



Tunneling

- Tunneling is used when the source and destination are on the same type of network, but there is a different network in between
- Source packets are encapsulated in packets, travelling through connecting network

Tunneling IPv6 Packets through IPv4

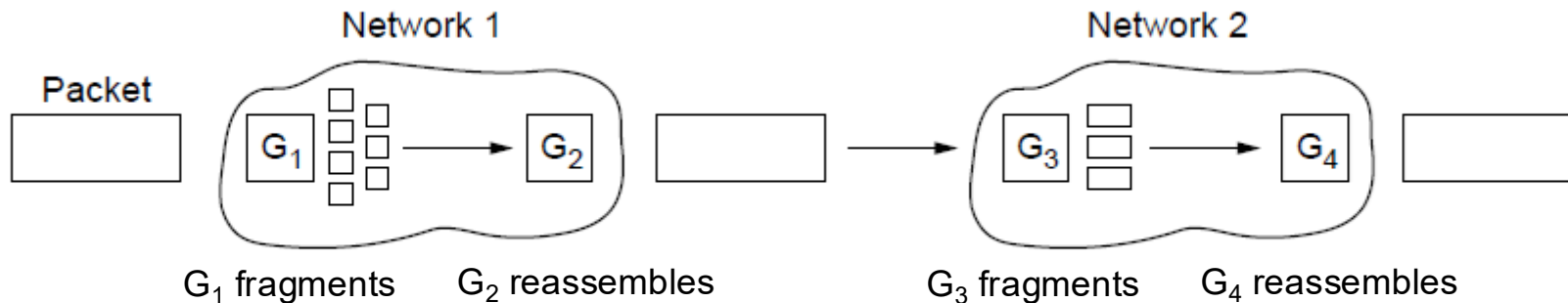


Fragmentation

- All networks define Maximum Transmission Unit (MTU), the largest packet size that can be transmitted
 - Hardware and operating system
 - Protocols and standards compliance
 - Efficiency of transmission
- Solution: fragmentation
 - Divides packets into fragments when routing large packets through a network with small MTU

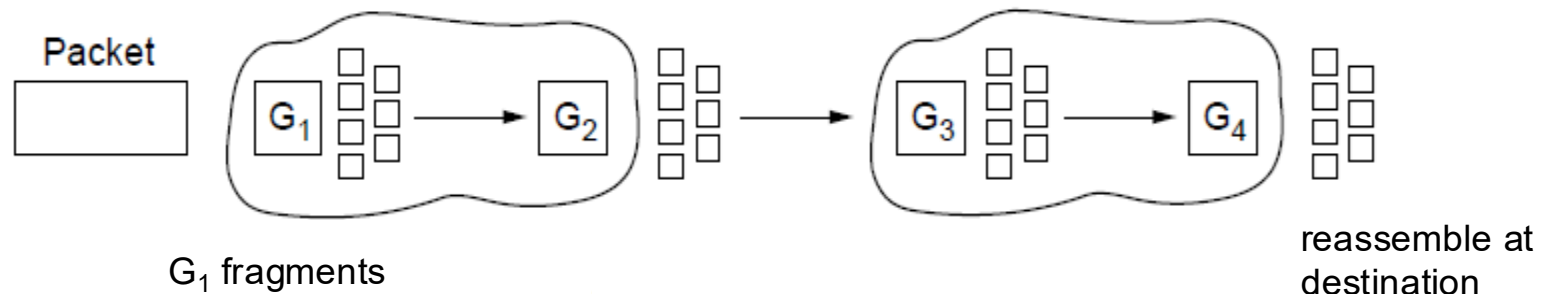
Types of Fragmentation (1)

- **Solution: Fragmentation and Reassembly**
- **Transparent:** packets fragmented & reassembled in each network. Route constrained.



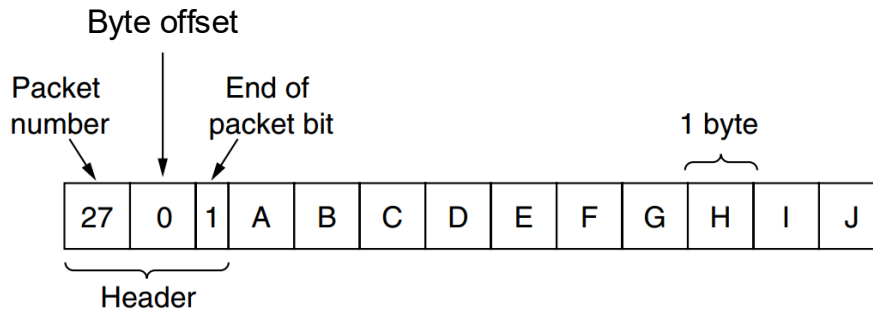
Types of Fragmentation (2)

- **Solution: Fragmentation and Reassembly**
- **Non-transparent:** fragments are reassembled at destination. IP uses this approach.
 - Router has less work.
 - Each packet requires packet number, byte offset, end of packet flag.



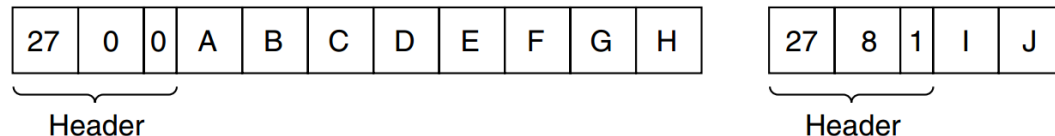
Example: IP-Style Fragmentation

Original packet:
(10 data bytes)



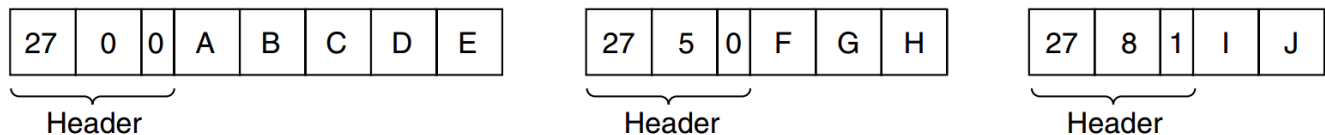
(a)

Fragmented:
(to 8 data bytes)



(b)

Re-fragmented:
(to 5 bytes)



(c)

Path MTU Discovery

- Alternative to fragmentation
- Advantage: the source knows the maximum packet length to send, no fragmentation in the network.
- If the routes and path MTU change, new error packets will be triggered, and the source must adapt to the updated path MTU

